Airline Passenger Demand Forecasting

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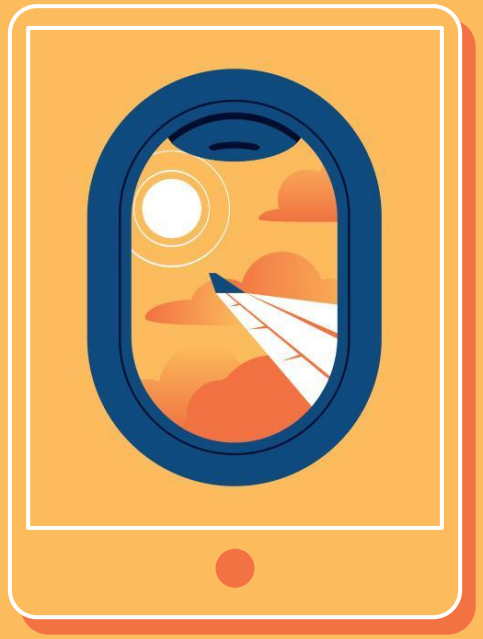
Problem Statement

- **Objective:**
  - To forecast total monthly **Revenue Passenger-Miles (RPM)** for the U.S. domestic market.
- **Target Variable:**
  - REV_PAX_MILES_140 (Revenue Passenger-Miles).
- **Why RPM?**
  - Simple "passenger counts" do not account for distance traveled.
  - **RPM = Paying Passengers × Miles Flown**. It is the most comprehensive metric for measuring airline economic activity and route performance.



Business Value

- **Stakeholders**:
  - Airlines, Airport Authorities, and Regulators.
- **Real-World Applications:**
  - **Capacity Planning**: Deciding fleet allocation (e.g., wide-body vs. narrow-body) to prevent empty seats or under-capacity.
  - **Dynamic Pricing**: Adjusting pricing strategies ahead of predicted seasonal peaks (e.g., summer).
  - **Infrastructure Ops**: Helping airports anticipate congestion and staffing needs.

Data Source & Combining Datasets

- U.S. DOT BTS – T-100 Domestic Segment dataset
- Includes data on time, airlines, regions, service type, demand, and capacity
- Combined 2022, 2023, 2024, and Jan–Jul 2025 datasets
- Final merged dataset: 26,306 rows × 37 columns

Data Cleaning & Preventing Leakage

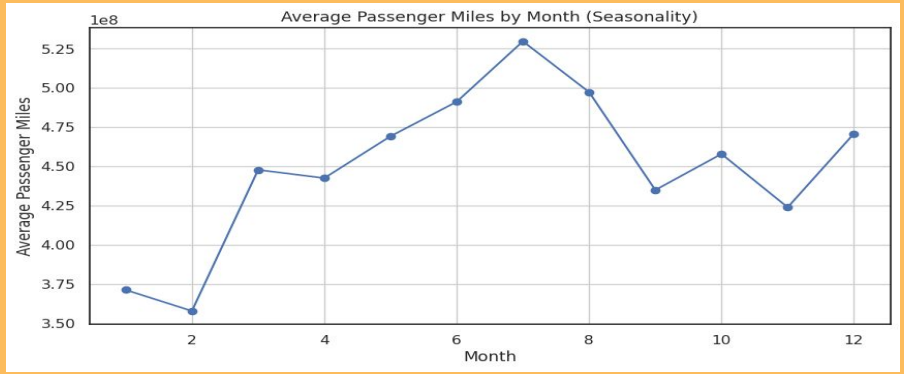
Standardized data types and removed columns with all zeros or missing values

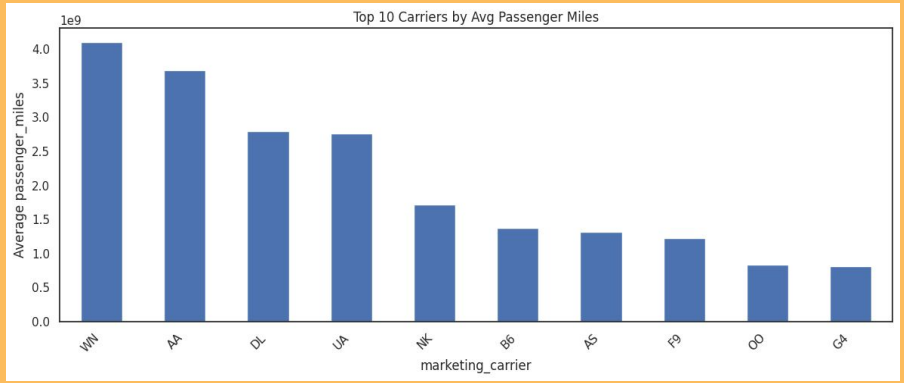
Dropped leakage variables that directly reflected the target

Removed cargo and mail fields unrelated to passenger demand

Renamed final set of 12 clean, and safe features for modeling

Key Exploratory Data Analysis





1. **Strong seasonality:** Passenger demand peaks in summer months (June–August) and is lowest in winter.
2. **Carrier dominance:** A few major airlines (WN, AA, DL, UA) generate the majority of passenger miles.
3. **Large variation across carriers:** Smaller regional airlines operate at much lower RPM levels, creating a highly skewed distribution.
4. **Consistent yearly trends:** Seasonal patterns repeat across years, indicating predictable demand cycles.

Modeling Framework

Models Applied

- Null Model
- Linear Regression
- Ridge Regression
- Decision Trees (Deep / Shallow / Random Search)
- Support Vector Regression (SVR)
  Random Forest (Baseline / Halving Grid / Random Search)
- Stacking Regressor

Reasoning

- Covers linear, non-linear, and ensemble modeling families.
  Enables comparison across complexity and generalization ability.

Linear Regression

- **Results:**
  - **RMSE**: 86M
  - **R^2**: 0.9972
- **Interpretation:**
  - RPM has a strong linear relationship with capacity variables.
  - Linear Regression already achieves near-perfect predictive power.
- **Ridge Regression**
  - Same RMSE and R^2 as Linear Regression.
  - Ridge $\alpha \approx 0 \Rightarrow$ Multicollinearity is not harming stability.

Model	CV RMSE	Test RMSE	Test R2
Null Model	~1.7 billion	~1.6 – 1.8 billion	~0.0
Linear Regression	93.7 million	86.7 million	0.9972
Ridge Regression	94.0 million	86.7 million	0.9972

Decision Tree

- **Deep Tree:**
  - **RMSE** ≈ 13M
  - **R^2** ≈ 0.9999
  - Severely overfits; memorizes data.
- **Shallow Tree:**
  - **RMSE** ≈ 512M
  - **R^2** ≈ 0.9010
  - Underfits; cannot capture linear structure.
- **Random Search Tree:**
  - RMSE ≈ 19M
  - **R^2** ≈ 0.9999
  - Better but still unstable and sensitive to sparse one-hot splits.

Model	CV RMSE	Test RMSE	R2 Score
Deep Tree	29,730,880	13,118,965	0.9999
Shallow Tree	458,743,913	512,168,871	0.9010
Random Tree	34,471,948	19,145,004	0.9999

Random Forest

- **Random Forest A (Baseline Model):**
  - **RMSE** ≈ 17.6M
  - **R^2** ≈ 0.9999
  - Best-performing single model.
- **Random Forest B (Halving Grid Search Tuned Model):**
  - **RMSE** ≈ 58M
  - **R^2** ≈ 0.9987
  - Over-regularized → performance dropped.
- **Random Forest C (Randomized Search Model):**
  - RMSE ≈ 42M
  - **R^2** ≈ 0.9993
  - Strong but not better than baseline.
- **Conclusion:**
  - RF captures nonlinear interactions without overfitting.
  - Most robust among individual models.

Model	Test RMSE	Test R2
RF A: Baseline	17,681,320	0.9999
RF B: Halving Grid	58,404,179	0.9987
RF C: Randomized	42,252,988	0.9993

Stacking Regressor

- **Components:**
  - **Linear Regression**
  - **Ridge Regression**
  - **Best Random Forest**
- **Meta-Learner:**
  - **Linear Regression**
- **Performance:**
  - RMSE ≈ 41.7M
  - **R^2** ≈ 0.9993
- **Why It Works:**
  - Linear models capture deterministic capacity-demand structure.
  - RF captures nonlinear carrier and region patterns.
  - Stacking blends strengths and reduces variance.

Challenges and Difficulties

Data Inconsistency and Cleaning

- Understand data

Feature Engineering and Leakage

- Leakage, $R^2 = 1$

Model Selection and Performance

- Multicollinearity
- SVR and Decision Tree

CONCLUSIONS

- **Model**
  - **Linear Regression & Ridge Regression:** Both linear models performed exceptionally well. This remarkable accuracy is attributed to the inherently linear relationship between operational capacity and passenger miles.
  - **Decision Trees:** Single Decision Trees proved unsuitable for this dataset. Rather overfit or underfit.
  - **Random Forest:** Random Forest performed very strongly. Halving Grid Search, Randomized Search slightly varied the performance, but it still perform well.
  - **Stacking Regressor:** The best-performing model was the Stacking Regressor, which combined Linear Regression, Ridge Regression, and the best Random Forest model.

- **Business Value**
  - **Optimized Capacity Planning:** Better allocation of fleet, routes, and crew.
  - **Enhanced Revenue Management:** Improved pricing strategies and more accurate revenue forecasts.
  - **Reduced Operational Costs:** Minimizing empty seats and unnecessary flights.

Thank You

Colab Link:
https://colab.research.google.com/drive/1mOlALlm0IkNLVVM53aOwmUaqVGP-4W4c?usp=sharing